



Mata Kuliah : Basis Data Lanjut  
Program Studi : D4 – Teknik Informatika  
Semester : 3

Kelas : 2E  
NIM : 254107020064  
Nama : Carlos Fadellilah Rama  
Jobsheet Ke- : 3

## Laporan Jobsheet

### 1. Praktikum 1

- Tampilkan 5 employee dengan gaji terendah

The screenshot shows the PostgreSQL Database Navigator on the left, displaying the 'company\_rama' database with a 'public' schema containing tables like 'employees' and 'projects'. The main window shows a SQL query in the 'Script-7' editor:

```
SELECT * FROM employees LIMIT 5;
```

The query results are displayed in a table with 5 rows and 2 columns: 'name' and 'salary'.

| name           | salary    |
|----------------|-----------|
| Fajar Nugroho  | 0         |
| Bambang Sutoyo | 6,500,000 |
| Siti Rahayu    | 7,500,000 |
| Lina Marlina   | 7,800,000 |
| Budi Santoso   | 8,000,000 |

- Tampilkan halaman 2 data projects (3 data per halaman)

The screenshot shows the PostgreSQL Database Navigator on the left, displaying the 'company\_rama' database with a 'public' schema containing tables like 'employees' and 'projects'. The main window shows a SQL query in the 'Script-7' editor:

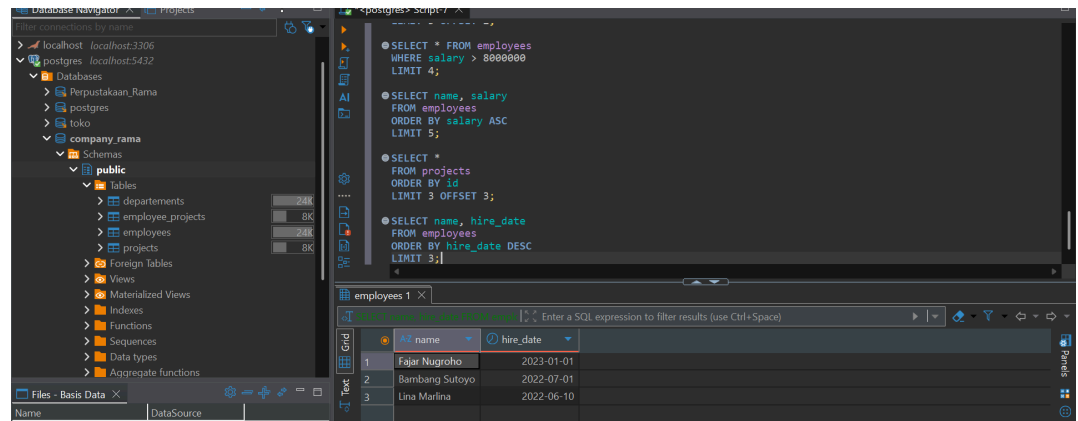
```
SELECT * FROM projects LIMIT 3 OFFSET 3;
```

The query results are displayed in a table with 3 rows and 4 columns: 'id', 'name', 'budget', and 'departement\_id'.

| id | name               | budget     | departement_id |
|----|--------------------|------------|----------------|
| 4  | Payroll System     | 45,000,000 | 3              |
| 5  | Marketing Campaign | 60,000,000 | 4              |
| 6  | Database Migration | 35,000,000 | 1              |

Menampilkan data proyek baris ke-4 hingga ke-6 (melewati 3 data pertama), yaitu Payroll System (id 4), Marketing Campaign (id 5), dan Database Migration (id 6).

- Tampilkan 3 employee yang dihire paling akhir



The screenshot shows the PostgreSQL IDE with a query window and a results grid. The query is as follows:

```
SELECT * FROM employees
WHERE salary > 8000000
LIMIT 4;

SELECT name, salary
FROM employees
ORDER BY salary ASC
LIMIT 5;

SELECT *
FROM projects
ORDER BY id
LIMIT 3 OFFSET 3;

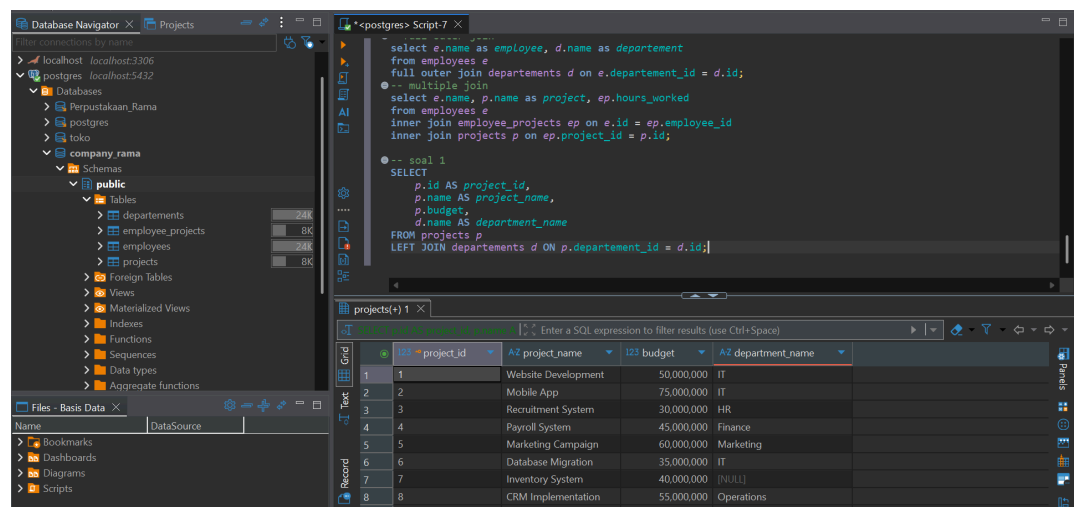
SELECT name, hire_date
FROM employees
ORDER BY hire_date DESC
LIMIT 3;
```

The results grid shows the last 3 employees hired, ordered by hire date in descending order:

| id | name           | hire_date  |
|----|----------------|------------|
| 1  | Fajar Nugroho  | 2023-01-01 |
| 2  | Bambang Sutuyo | 2022-07-01 |
| 3  | Lina Marlina   | 2022-06-10 |

## 2. Praktikum 2

- Tampilkan semua projects beserta department penanggung jawabnya



The screenshot shows the PostgreSQL IDE with a query window and a results grid. The query is as follows:

```
select e.name as employee, d.name as department
from employees e
full outer join departments d on e.department_id = d.id;

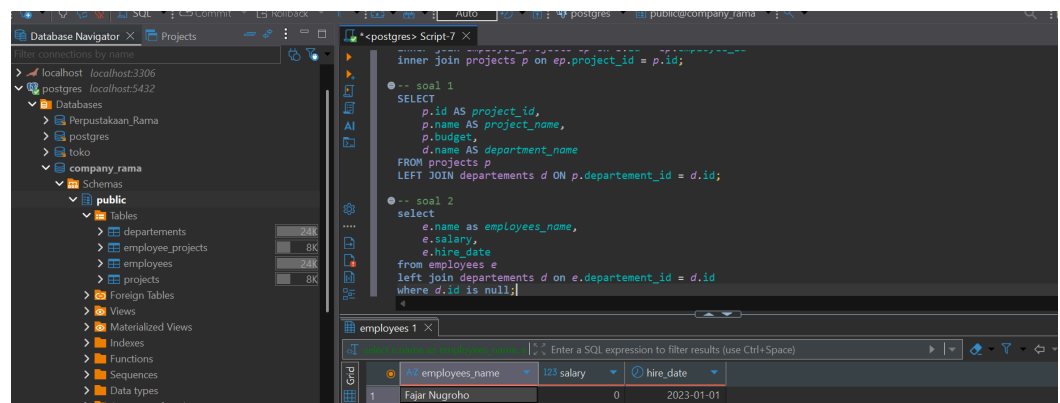
-- multiple join
select e.name, p.name as project, ep.hours_worked
from employees e
inner join employee_projects ep on e.id = ep.employee_id
inner join projects p on ep.project_id = p.id;

-- soal 1
SELECT
p.id AS project_id,
p.name AS project_name,
p.budget,
d.name AS department_name
FROM projects p
LEFT JOIN departments d ON p.department_id = d.id;
```

The results grid shows the projects and their departments:

| id | project_id | project_name        | budget     | department_name |
|----|------------|---------------------|------------|-----------------|
| 1  | 1          | Website Development | 50,000,000 | IT              |
| 2  | 2          | Mobile App          | 75,000,000 | IT              |
| 3  | 3          | Recruitment System  | 30,000,000 | HR              |
| 4  | 4          | Payroll System      | 45,000,000 | Finance         |
| 5  | 5          | Marketing Campaign  | 60,000,000 | Marketing       |
| 6  | 6          | Database Migration  | 35,000,000 | IT              |
| 7  | 7          | Inventory System    | 40,000,000 | NULL            |
| 8  | 8          | CRM Implementation  | 55,000,000 | Operations      |

- Tampilkan employee yang tidak memiliki department



The screenshot shows the PostgreSQL IDE with a query window and a results grid. The query is as follows:

```
inner join projects p on ep.project_id = p.id;

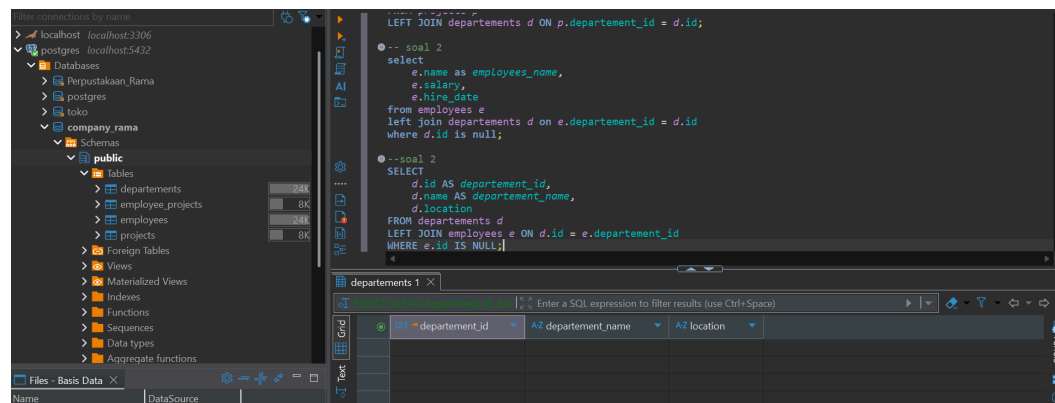
-- soal 1
SELECT
p.id AS project_id,
p.name AS project_name,
p.budget,
d.name AS department_name
FROM projects p
LEFT JOIN departments d ON p.department_id = d.id;

-- soal 2
select
e.name as employees_name,
e.salary,
e.hire_date
from employees e
left join departments d on e.department_id = d.id
where d.id is null;
```

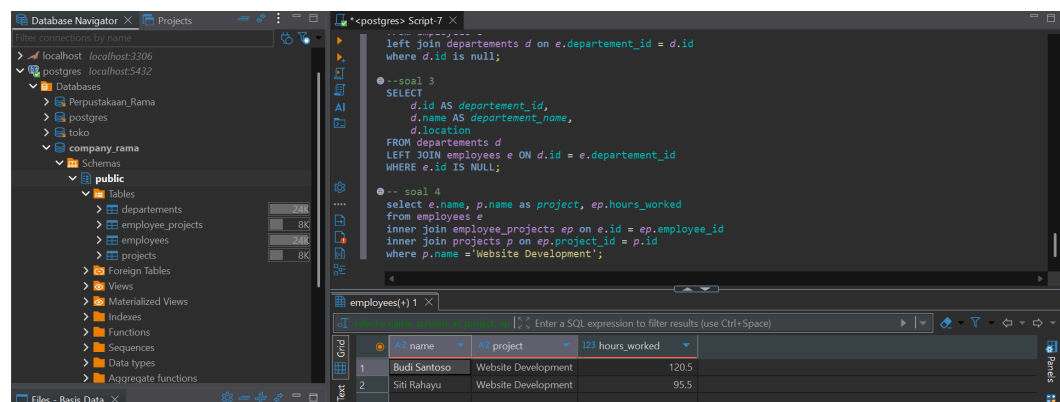
The results grid shows the employees without a department:

| id | employees_name | salary | hire_date  |
|----|----------------|--------|------------|
| 1  | Fajar Nugroho  | 0      | 2023-01-01 |

- Tampilkan department yang tidak memiliki employee

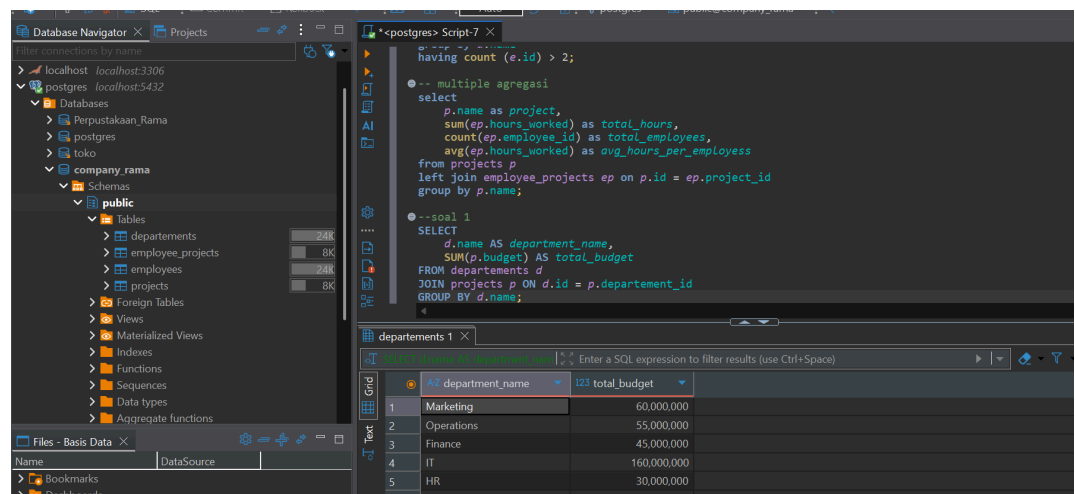


- Tampilkan employee yang bekerja di project 'Website Development'

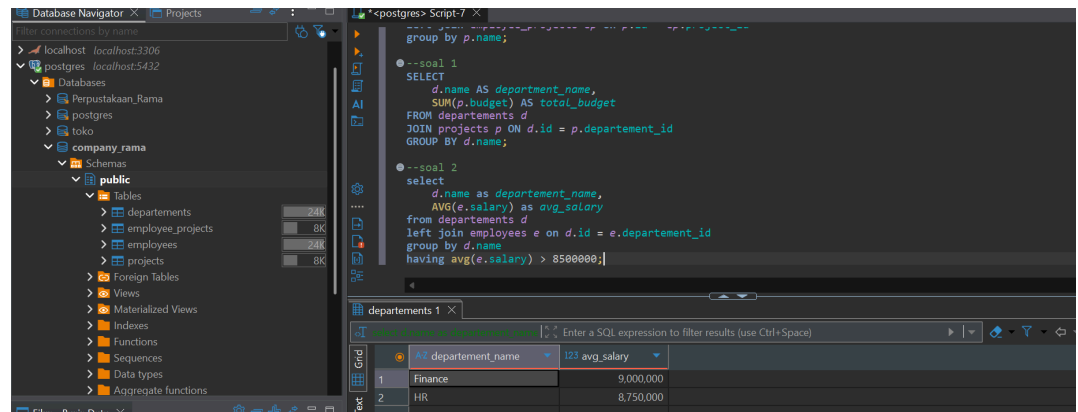


### 3. Praktikum 3

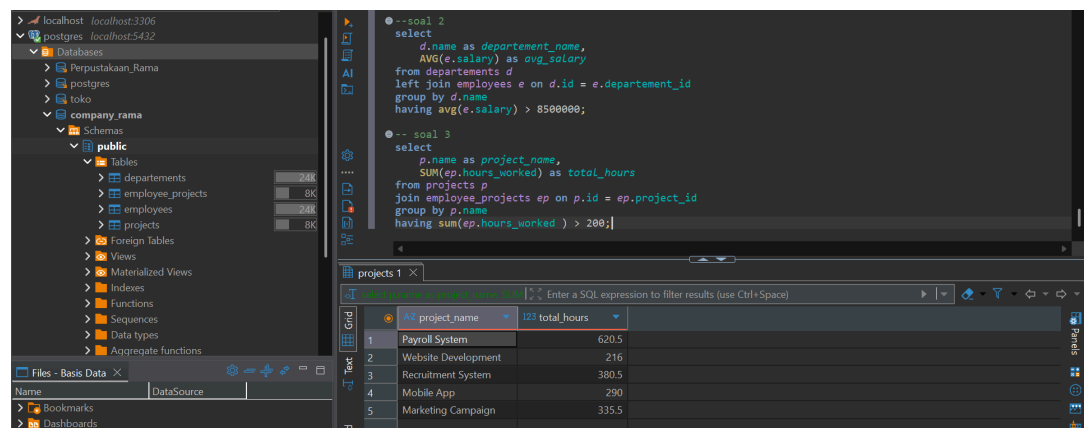
- Tampilkan total budget projects per department



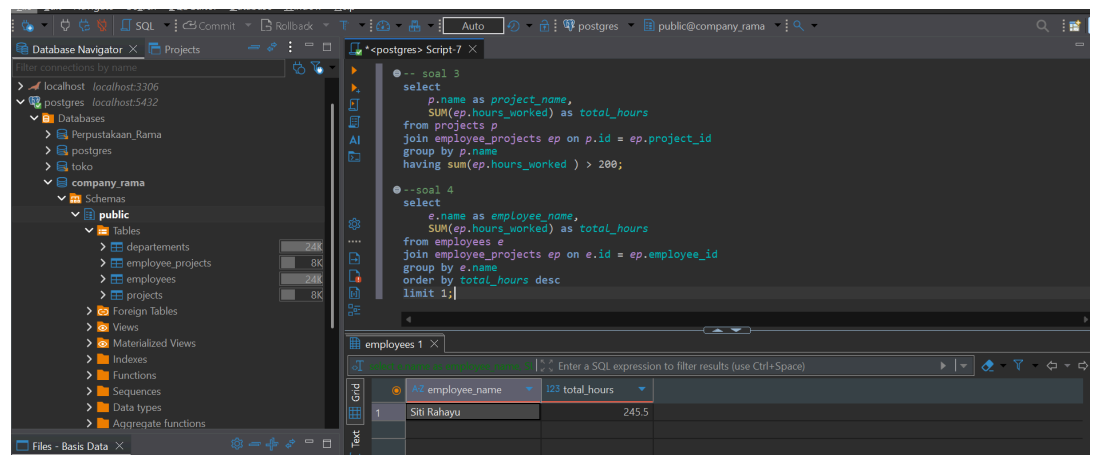
- Tampilkan department dengan rata-rata gaji di atas 8.5 juta



- Tampilkan project dengan total jam kerja lebih dari 200 jam

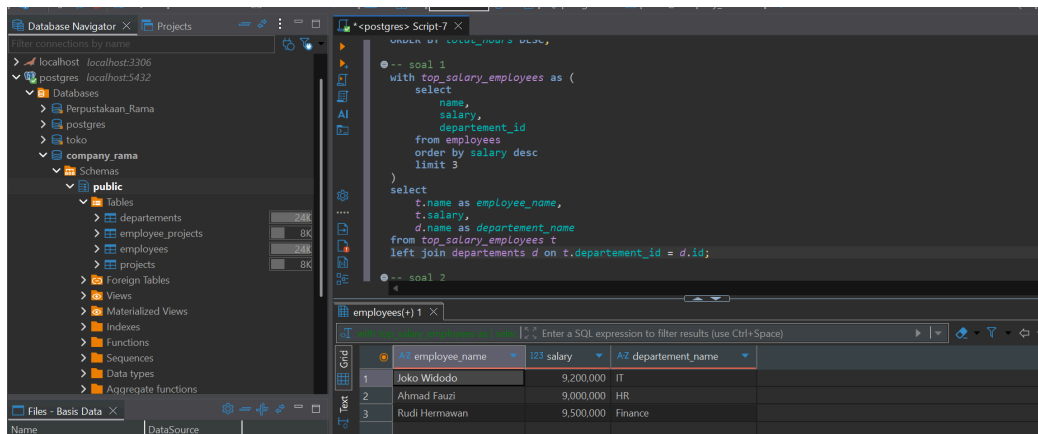


- Tampilkan employee dengan total jam kerja terbanyak

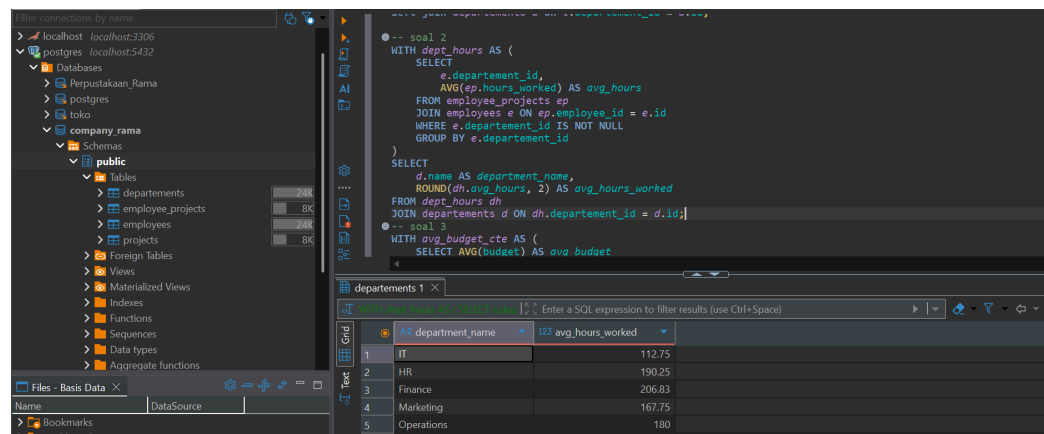


#### 4. Praktikum 4

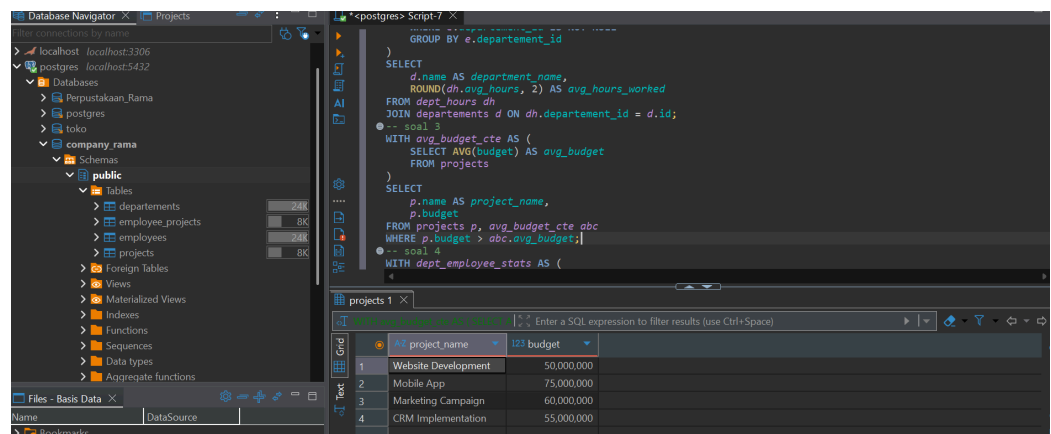
- Buat CTE untuk menampilkan top 3 employee dengan gaji tertinggi



- Buat CTE untuk menghitung rata-rata jam kerja per department



- Buat CTE untuk menampilkan project dengan budget di atas rata-rata



- Buat CTE berjenjang (multiple CTEs) untuk analisis department lengkap

Database Navigator

Projects

View connections by name

localhost:3306

postgres:localhost:5432

Databases

Perpustakaan\_Rama

postgres

toko

company\_rama

Schemas

public

Tables

departements24K

employee\_projects8K

employees24K

projects8K

Foreign Tables

Views

Materialized Views

Indexes

Functions

Sequences

Data types

Aggregate functions

Files - Basic Data

NameDataSource

Bookmarks

Dashboards

postgres> Script-7

```
dept_project_stats AS (  
  SELECT  
    departement_id,  
    COUNT(id) AS total_projects,  
    SUM(budget) AS total_budget  
  FROM projects  
  WHERE departement_id IS NOT NULL  
  GROUP BY departement_id  
)  
SELECT  
  d.name AS department_name,  
  COALESCE(es.total_employees, 0) AS total_employees,  
  COALESCE(ROUND(es.avg_salary, 2), 0) AS avg_salary,  
  COALESCE(ps.total_projects, 0) AS total_projects,  
  COALESCE(ps.total_budget, 0) AS total_budget  
FROM departements d  
LEFT JOIN dept_employee_stats es ON d.id = es.departement_id  
LEFT JOIN dept_project_stats ps ON d.id = ps.departement_id;
```

departments 1

WITH dept\_employee\_stats AS (SELECT

Enter a SQL expression to filter results (use Ctrl+Space)

|   | department_name | total_employees | avg_salary   | total_projects | total_budget |
|---|-----------------|-----------------|--------------|----------------|--------------|
| 1 | IT              | 3               | 8,233.333.33 | 3              | 160,000,000  |
| 2 | HR              | 2               | 8,750,000    | 1              | 30,000,000   |
| 3 | Finance         | 3               | 9,000,000    | 1              | 45,000,000   |
| 4 | Marketing       | 2               | 8,000,000    | 1              | 60,000,000   |
| 5 | Operations      | 1               | 6,500,000    | 1              | 55,000,000   |